MICRO-GEOM errors

Surface Finish

Small variation that we can
Have on the surface

Groover M.P.: Chapters 6 and 20

Learning Targets

The student knows:

- what surface finish and integrity are;
- how to measure the surface finish (roughness and waviness);
- how to predict the surface finish due to a machining process.

surface quality is a big issue.. roughness / waviness

↓

different kind of geometry error

you can have different geometry issue but also others problem about surface

All possible deviation with respect to ideal shape that affect product performance

WHY ARE SURFACES IMPORTANT?

- Aesthetic reasons
- Surfaces affect safety
- Friction and wear depend on surface characteristics
- Surfaces affect mechanical and physical properties
- Assembly of parts is affected by their surfaces
- Smooth surfaces make better electrical contacts

[Property of surface]
relevant roughness of surface to affect the reflection property +
FRICTION, WEAR influenced by SURFACE quality

on FATIGUE issue
small surface error can be an initial crack propagation
⇓
object Break!
↑ weak point of the obj... subject to periodic cycle of load (mech/termal)

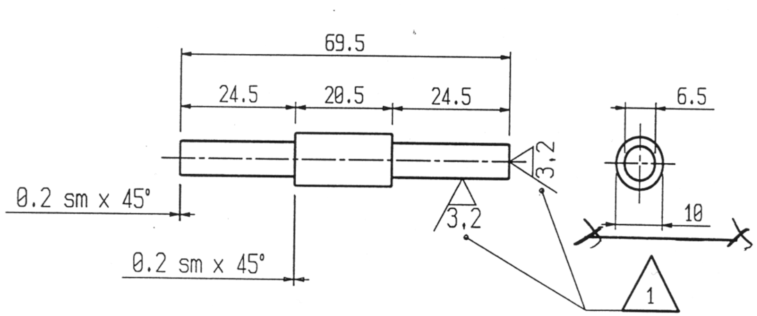
(ART element)

Cloud gate (A. Kapoor, 2006)
Millennium Park - Chicago
Stainless Steel

we need to define the requirements → by understanding the kind of error! ⇒

Ideal Surface: Designer's intended surface contour of part, defined by lines in the engineering drawing

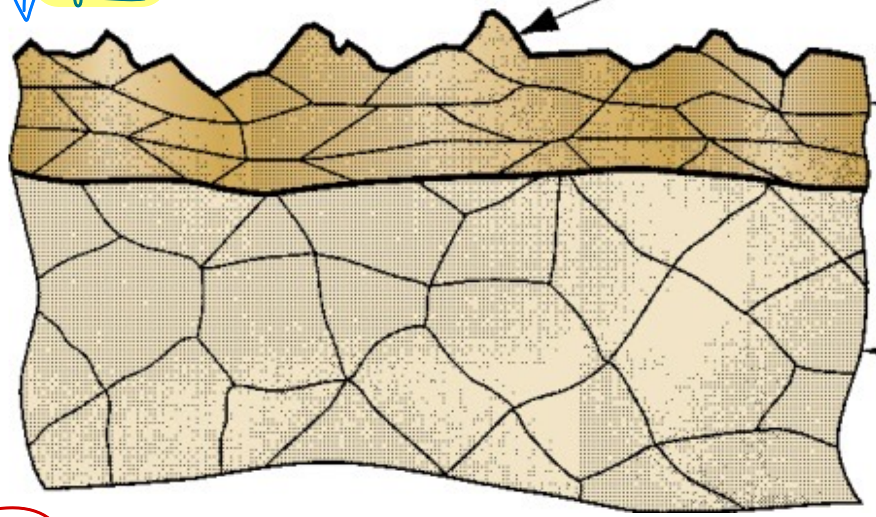
- Ideal surfaces appear as absolutely straight lines, ideal circles, round holes, and other edges and surfaces that are geometrically perfect
- Actual surfaces of a part are determined by the manufacturing processes used to make them
- Variety of processes result in wide variations in surface characteristics



"90034" design Christopher Dresser, 1885 surface after processing (geometrical variation)

Magnified cross section of a typical metallic part surface

during machining layer with defects



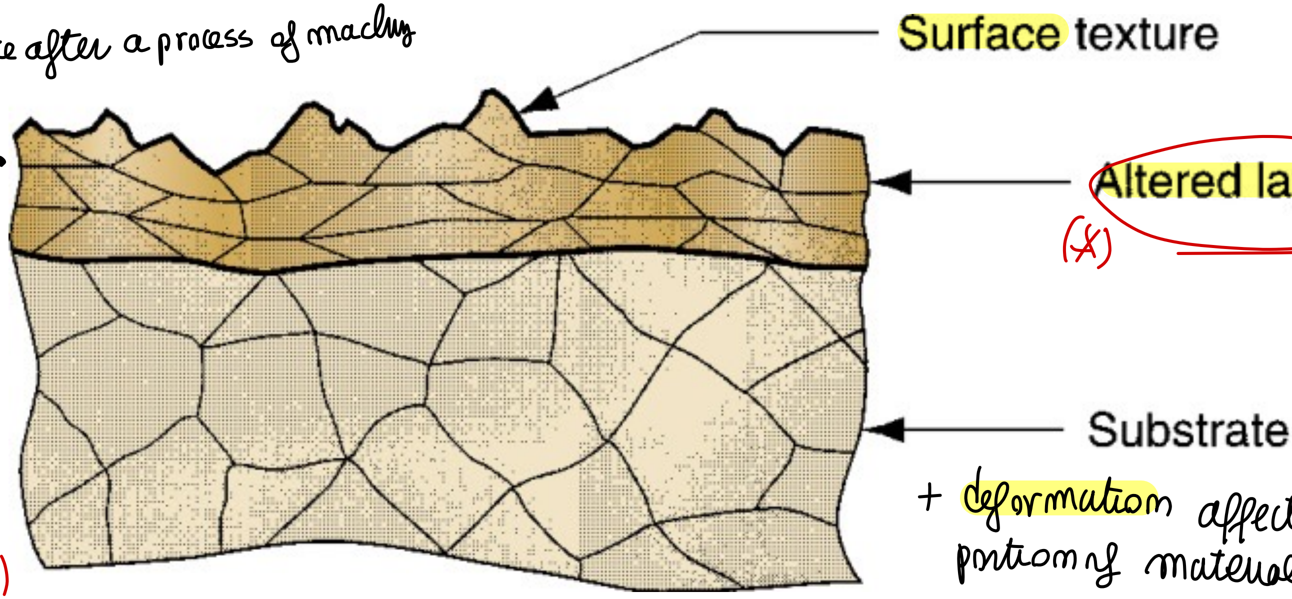
Forces, Heat during machining → changing property

Thermal cycles changing crystals of my product... interaction = changes on material + Deformation

Magnified cross section of a typical metallic part surface

surface after a process of machining

{geometrical variation} →



(*)

Machining ↓
layer of material surface with differences in the geometry and property (*)

+ deformation affects a portion of material

SURFACE finish: there are more mechanical variation
during machining we have heat that changes the property during thermal cycle (heat/cool) → affect object grains: interactions that change the material (limited to the material affected to cycle)

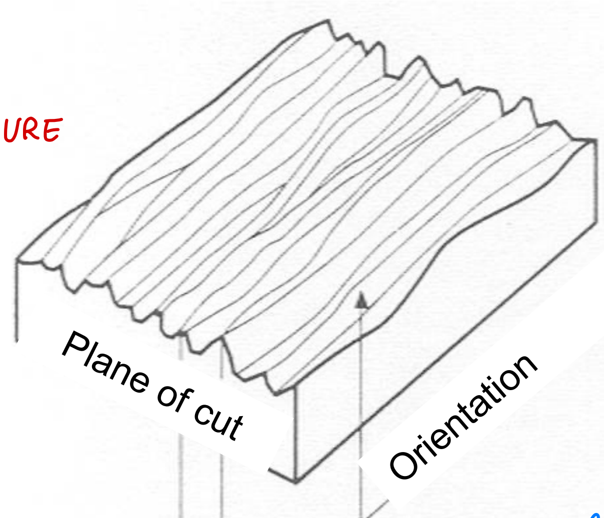
SURFACE TEXTURE

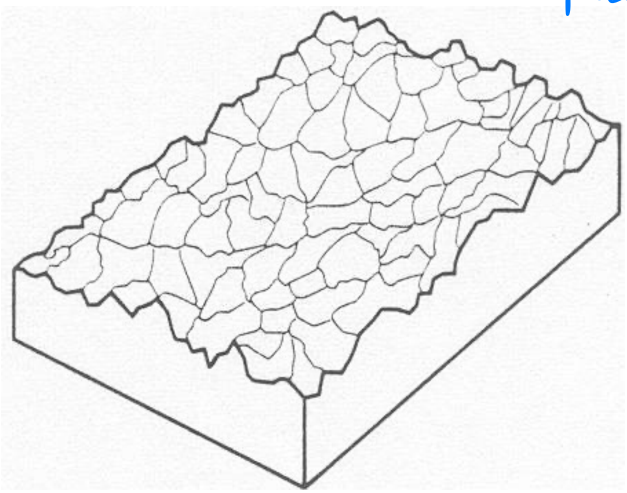
Geometry: NOT ideal smooth surface but a more complex shape

The topography and geometric features of the surface

- When highly magnified, the surface is anything but straight and smooth. It has roughness, waviness, and flaws. (RANDOM irregularities or ORIENTED)
  ↳ with a given pattern depending on the
- It also possesses a pattern and/or direction resulting from the mechanical process that produced it. type of machining process

surface errors
↓
SURFACE TEXTURE ERROR ⇒



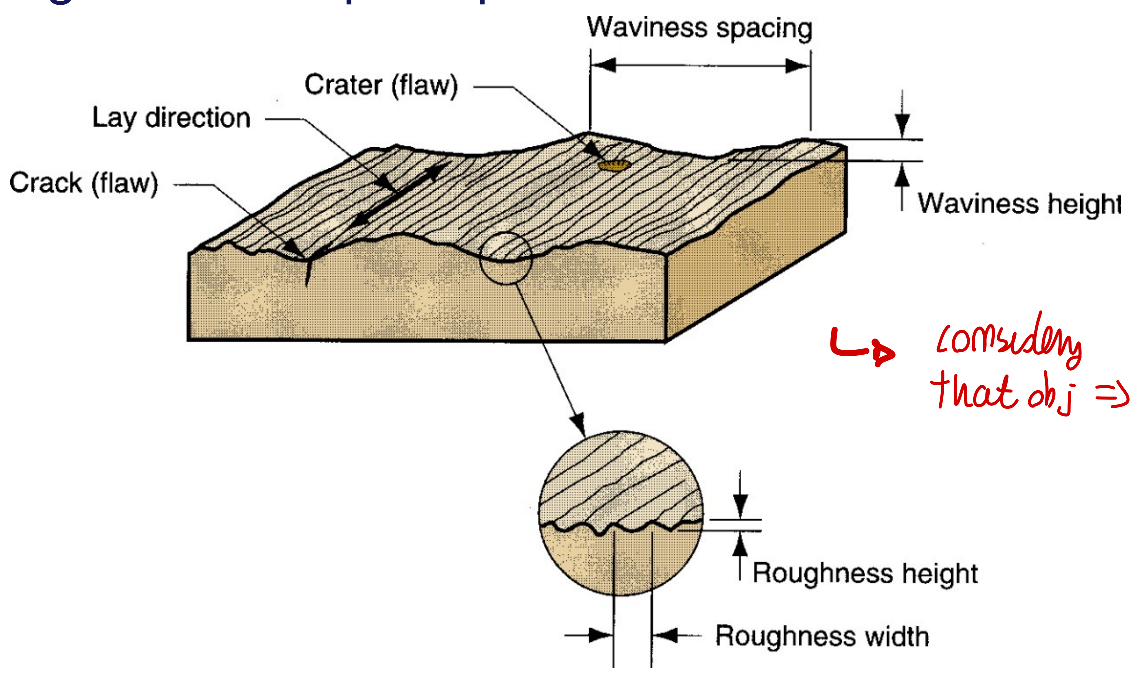


Oriented ← possible surface irregularities → Random

FOUR ELEMENTS OF SURFACE TEXTURE

(Possible ERRORS) MICRO ERRORS

1. Roughness: small, finely-spaced deviations from ideal surface
   - Determined by material characteristics and processes that formed the surface.
2. Waviness: deviations of much larger spacing
   - Waviness deviations occur due to work deflection, vibration, tooling, and similar factors,
   - Roughness is superimposed on waviness.



considery that obj ⇒

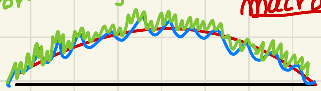
after machining

FORM error

(*) ROUGHNESS error

macro (FORM) error ← geometrical error

{ we have source of geom. error }



ideal line surface

real surface final ↓ due to

- FORM ERRORS (MACRO Geom)
- Waveness
- Roughness

small variation superimposed to waviness (*)

WAVENESS

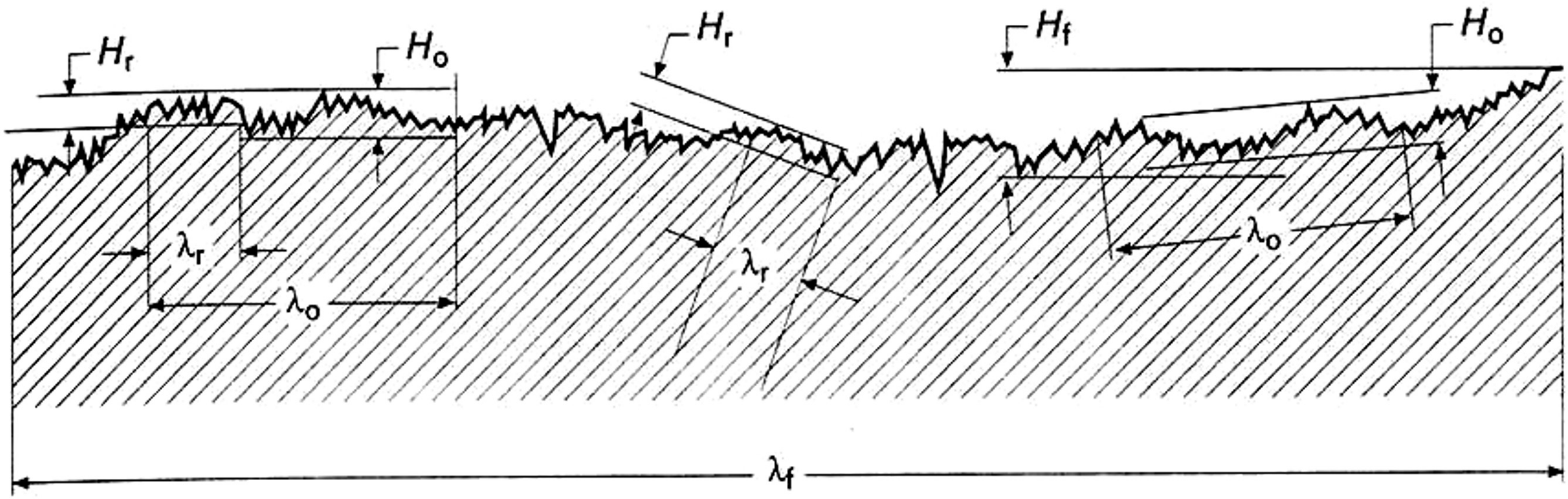
Interaction of tool with material, causing VIBRATIONS → super-imposed errors respect to the MACRO GEOM error

(given frequency error)

FOUR ELEMENTS OF SURFACE TEXTURE

to distinguish all the ERRORS

we collect different points → FILTERING the signal



error classif.

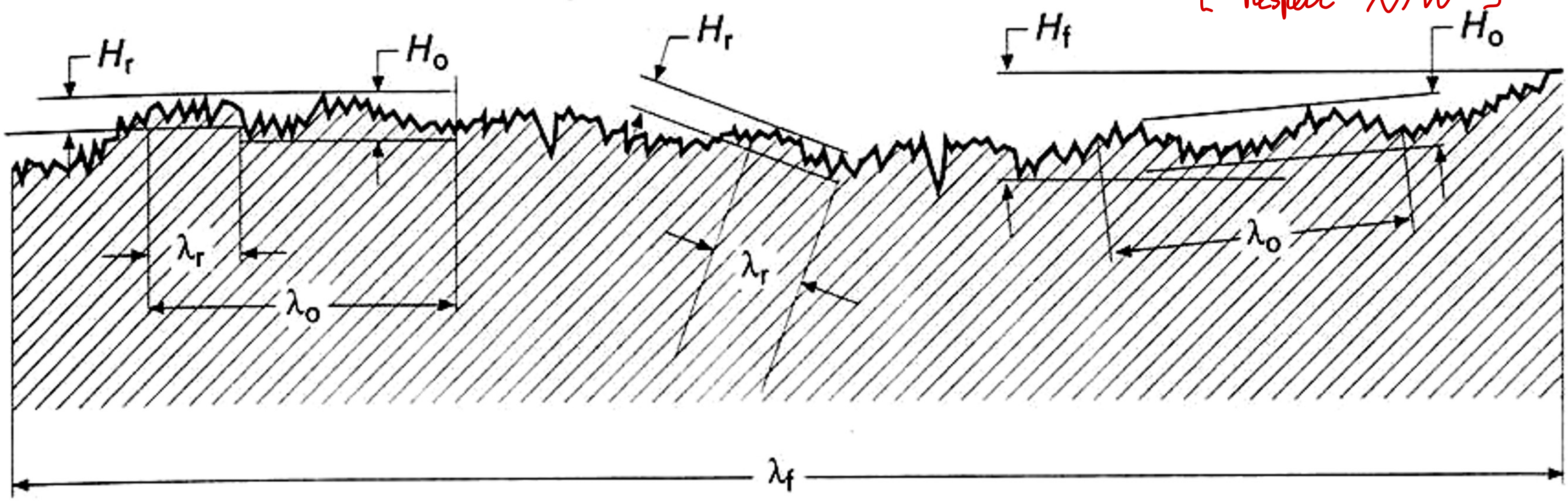
Type of deviation		Ratio wavelength/height
Microgeometric deviations	Roughness or primary texture	$0 < \lambda_r/h_r \leq 50$
	VIBRATIONS related! → tool shape Waviness or secondary texture	$50 < \lambda_o/h_o \leq 1000$
Macrogeometric deviations	Form errors	$1000 < \lambda_f/h_f$

through machine design we should limit vibration (mechanical)

FOUR ELEMENTS OF SURFACE TEXTURE

To classify the error, if the ratio between wave length / height is below 50 (*)

[CLASSIFY ERROR Respect λ/h]



waviness caused by machine ↓ its design must limit mechanical system that should limit VIBRATION!

Type of deviation		Ratio wavelength/height
Microgeometric deviations	Roughness or primary texture	$0 < \lambda_r / h_r \leq 50$
	Waviness or secondary texture	$50 < \lambda_o / h_o \leq 1000$
Macrogeometric deviations	Form errors	$1000 < \lambda_f / h_f$

(*)

→ related to VIBRATIONS! due to tool shape and machine

Roughness → depends more on process parameter and interaction TOOL - MATERIAL (WE FOCUS ON ROUGHNESS)

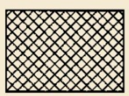
FOUR ELEMENTS OF SURFACE TEXTURE

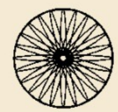
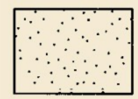
when speaking about surface texture..

PATTERN that characterize the small surface error

3. Lay - predominant direction or pattern of the surface texture

to characterize the pattern, direction of surface predominant

Lay symbol	Surface pattern	Description
=		Lay is parallel to line representing surface to which symbol is applied.
⊥		Lay is perpendicular to line representing surface to which symbol is applied.
X		Lay is angular in both directions to line representing surface to which symbol is applied.

Lay symbol	Surface pattern	Description
C		Lay is circular relative to center of surface to which symbol is applied.
R		Lay is approximately radial relative to the center of the surface to which symbol is applied.
P		Lay is particulate, nondirectional, or protuberant.

{or just SCRATCH (LOCAL DEFECTS, irregularities) CRACK, IMPURITIES}

fracture → surface inside, impurity

called FLAWS

4. Flaws - irregularities that occur occasionally on the surface

- Includes cracks, scratches, inclusions, and similar defects in the surface
- Although some flaws relate to surface texture, they also affect surface integrity

Ex of IMPURITIES: Built-up-Edge, portion of material with impurity inside surface

all this, characterize surface TEXTURE

SURFACE ROUGHNESS AND SURFACE FINISH

Surface roughness: a measurable characteristic based on roughness deviations

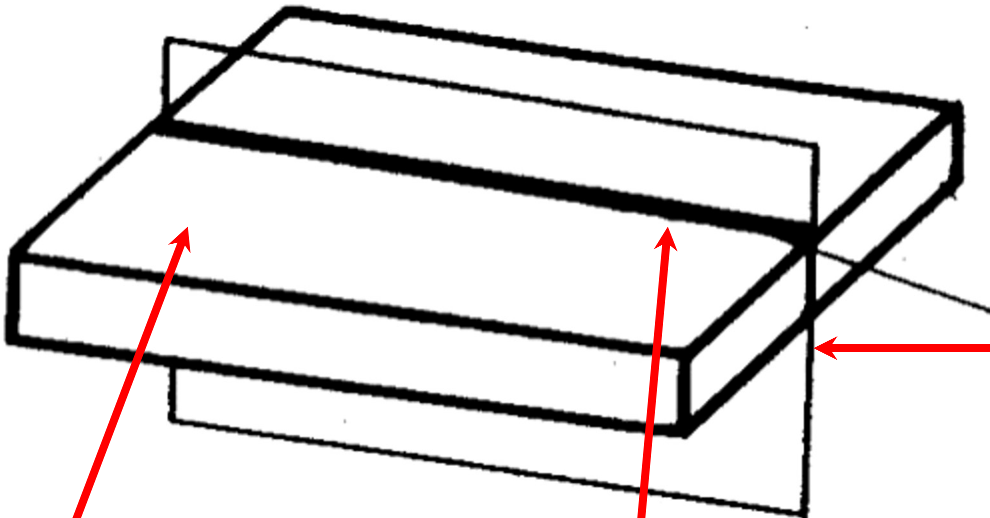
We concentrate on Roughness

Surface finish: a more subjective term denoting smoothness and general quality of a surface

- In popular usage, surface finish is often used as a synonym for surface roughness
- Both terms are within the scope of surface texture

finish is related to everything → also mechanical verification of external layer ...
general properties of surface (ALL!)

To understand and characterize ROUGHNESS

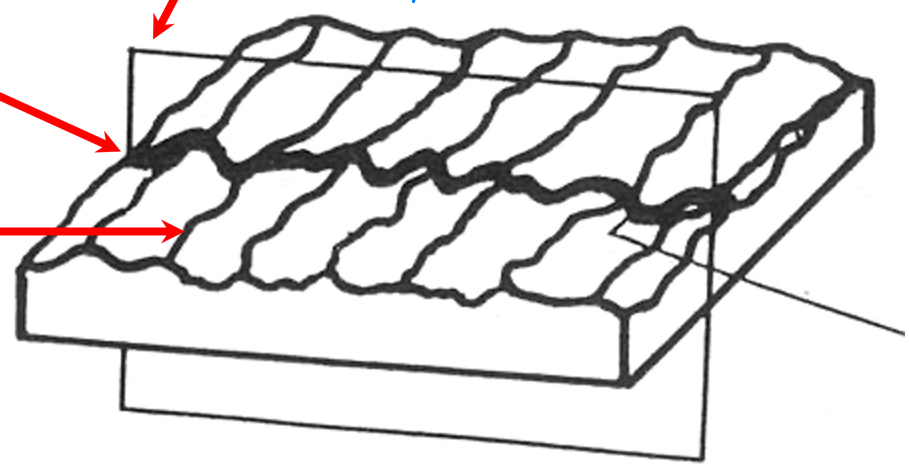


Intersection plane

Ideal surface

Ideal profile

super imposition of 3 errors



Real profile

after cutting --

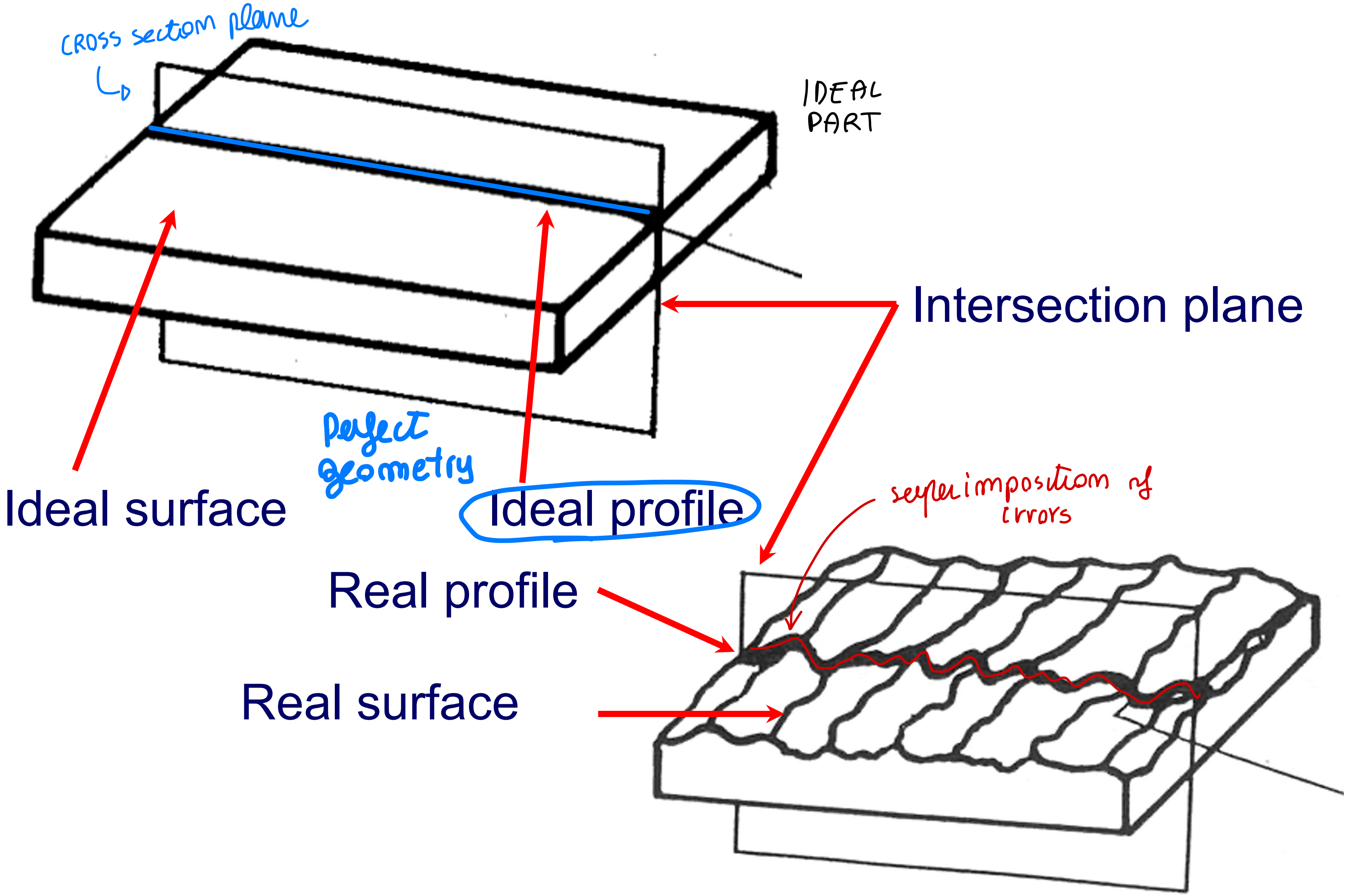
NO PERFECT geometry

Real surface

different profile than expected

SURFACE TEXTURE

How to characterize Roughness



WE analyze

SURFACE TEXTURE

surface finish: Related to multiple characteristic

The following surfaces must be considered in the study of the surface texture: Different kind of surface

- **Ideal surface**: surface as designed by the designer → perfect smooth
- **Real surface**: surface limiting the body and separating it from the surrounding medium (it is generated by the manufacturing process)

  OUTPUT OF PROCESS
- **Primary Surface**: Surface probed by means of a measuring system
- **Reference surface**: Surface with reference to which the real surface is measured
- **Mean surface**: Surface identical to the reference surface, but located as resulting from a least squares fitting

able to measure .. Roughness, Waviness → can be measured!

MEASUREMENT OF SURFACE ROUGHNESS

Three methods to measure surface roughness:

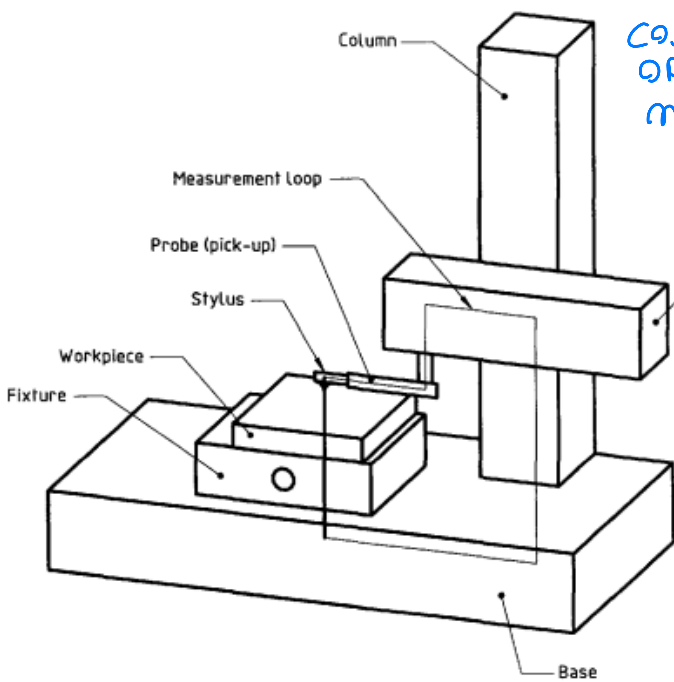
{ Measuring device to evaluate Roughness }

measurement device of Roughness ⟹

1. Subjective comparison with standard test surfaces
   - Fingernail test
2. **Stylus electronic instruments**
3. Optical techniques

small STYLUS
↱ Interacting with the surface.. NOT able to collect on REAL surface → mechanical filter

(move the nail on the part to evaluate quality) → bad estimation... measurement

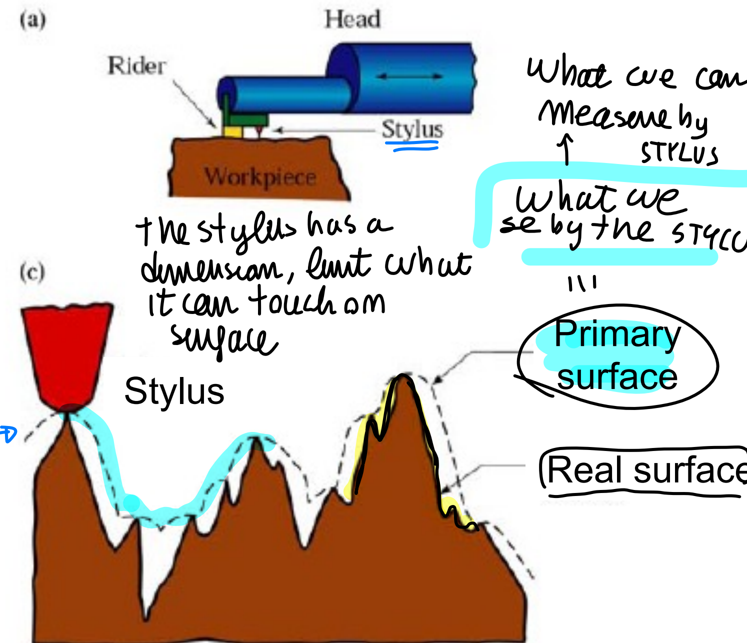


CONTACT or OPTICAL methods

evaluate WAVINESS. ROUGHNESS from this measurement

⇑

Primary surface is what we have to evaluate surface property!



the stylus has a dimension, limit what it can touch on surface

What we can measure by STYLUS ↑

what we see by the STYLUS

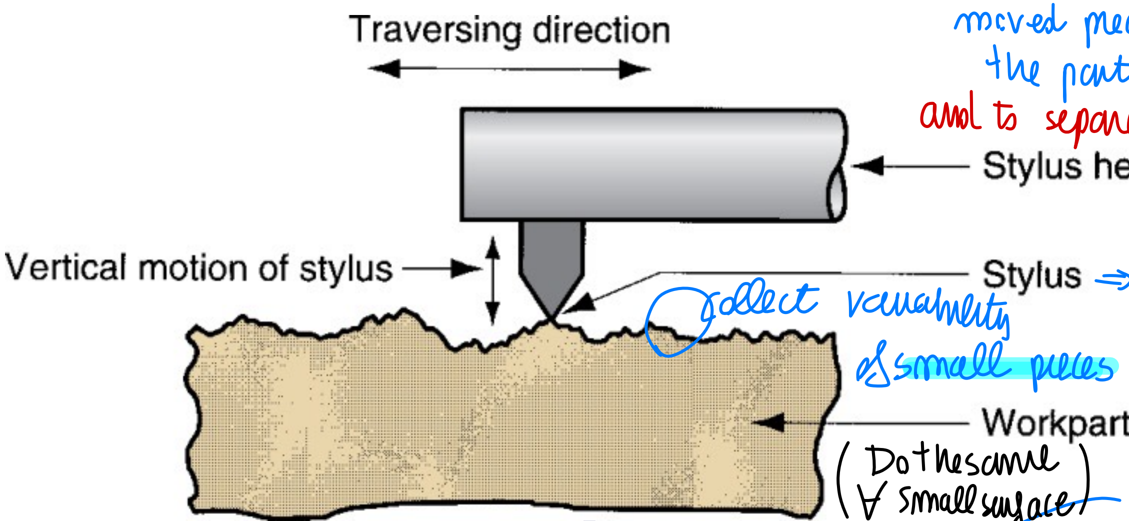
III

STYLUS dimension → mechanical filter

Measurement of Surface Roughness

Stylus head traverses horizontally across surface, while stylus moves vertically to follow surface profile.

through the small tip stylus, move along a linear path trajectory
moved precisely, resisting deviation along the part → COLLECTING all values of surface
and to separate the different possible errors we should intelligent collect POINTS and FILTERING the signal



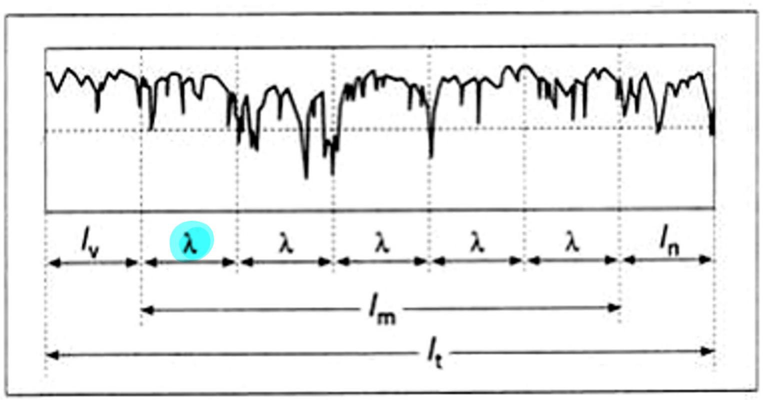
collect variability of small pieces

Stylus ⇒ move along a linear path
RESISTING deviation

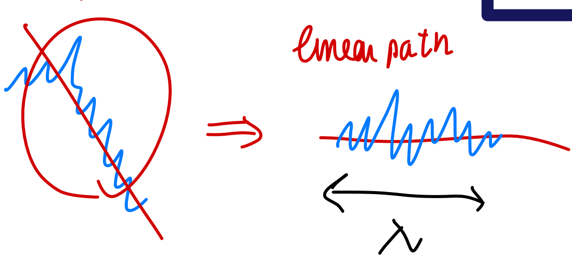
→ to filter the point.. we usually consider a small portion of the surface.. collecting variability and linearize variability

(Do the same ∀ small surface)

Linear path



we collect different information!
(slope + variation)



TRASFORM VARIABILITY
Remove slope and linearize

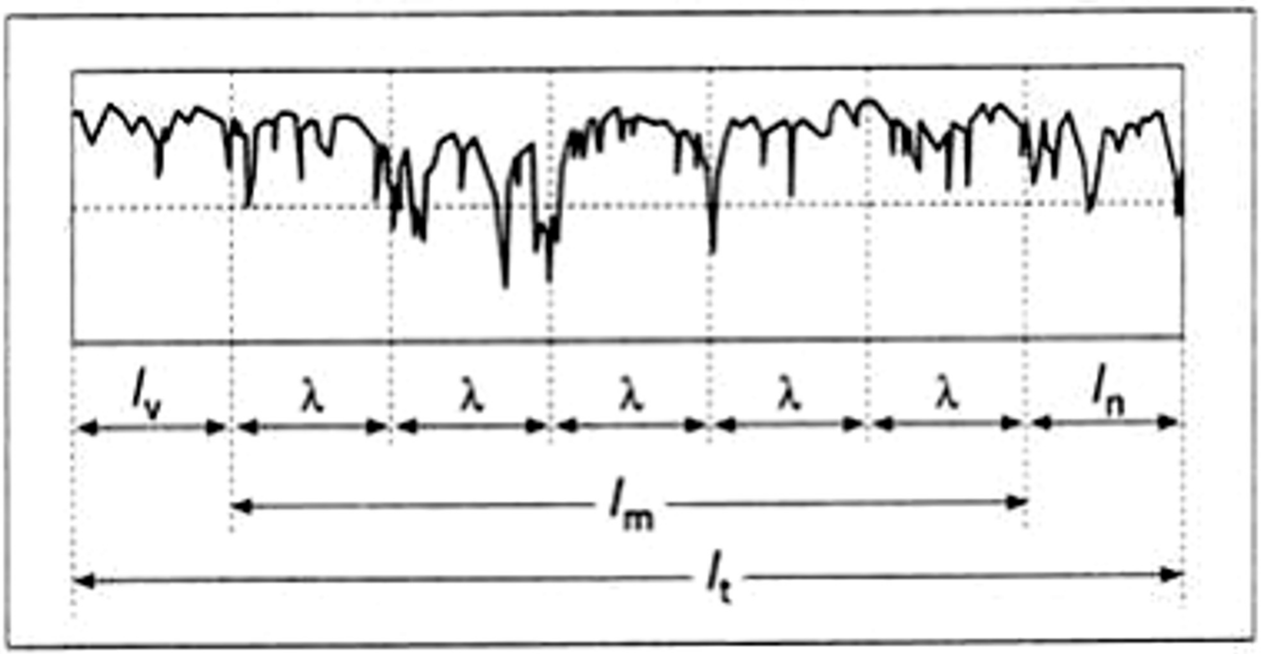
Cutoff Length

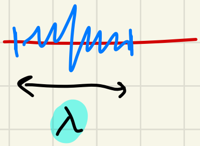
A problem with the R_a computation is that waviness may get included.

To deal with this problem, a parameter called the **cutoff length** is used as a filter to separate waviness from roughness deviations.

Cutoff length is a sampling distance along the surface:

- A sampling distance shorter than the waviness eliminates waviness deviations and only includes roughness deviations.

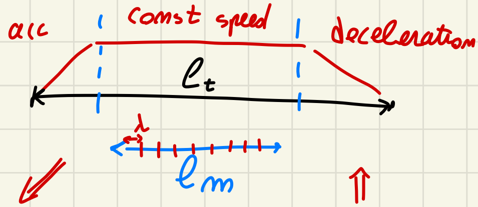




consider CUT-OFF lenght → to obtain a small piece of surface at once

small lenght to filter a partial frequency Information {use that INFO}

we consider a portion of total lenght (consider a small segment λ, we divide in small pieces λ obtaing a total profile of surface variability)



[you have to chose cut off lenght λ] respect the expected Roughness

↳ to evaluate $\boxed{\text{ROUGHNESS}}$

We consider a portion with constant velocity

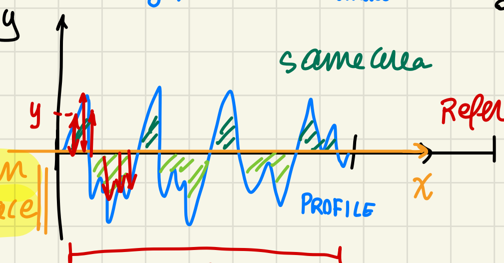
When considering the real lenght of measuring, so the usable data for evaluation, IT IS JUST A portion of the total lenght ℓ_T that stylus travel

{because acc, dec influence the measurements...}

so we consider JUST the portion $\ell_m < \ell_t$, we must be on same measuring condition ∀ point (const speed) (NO DYNAMIC variability) WE Require const velocity

once measured ↓ to evaluate ROUGHNESS

you will have a profile coming from measurement



mean surface ↓ respect which you evaluate deviation (y)

↓ To position the axis of the reference y ado it to have a positioned surface (origin where you start to measure and x ≃ mean surface)

as same area up/down the x axis ← to minimize least square error!

Reference x so that the area above/below x is the same ← x positioned as a Reference

(LEAST SQUARE MINIMIZATION of DEVIATION) ← OUTPUT of

ROUGHNESS → related to more parameters! different kind of parameter ROUGHNESS {(PROFILE / AREAL Roughness)

PARAMETERS representing DEVIATION (height) / SPACING (WAVE lenght), ecc-- surface property

deviation → average ROUGHNESS :

measurement lenght — absolute deviation value $|y|$

$$R_a = \frac{1}{\ell_m} \int_0^{\ell_m} |y| \, dx$$

{arithmetic average error} ["average" respect ℓ_m]

discrete profile measurement!

using the INFORMATION about continuous profile y

When speaking of INSTRUMENTS → we have collection of POINTS on the profile..

DISCRETE PROFILE measurement

consider discrete sampling of profile

m points

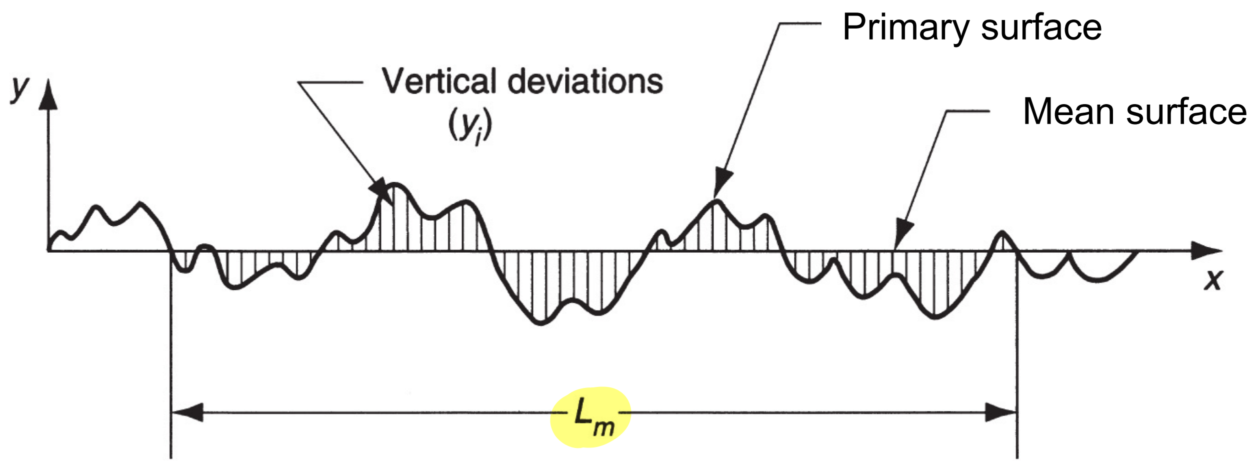
$$\boxed{R_a = \frac{1}{m}\sum_{i=1}^{m} |y_i|}$$

typical

- most used parameters!

SURFACE ROUGHNESS

Vertical deviations from mean surface over a specified surface length



Arithmetic average based on absolute values of deviations, and it is referred to as *average roughness* (R_a)

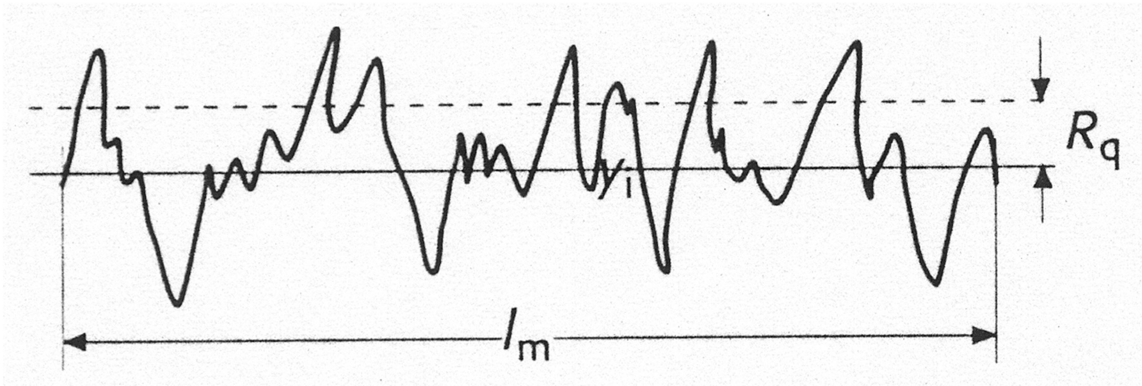
$$R_a = \int_0^{Lm} \frac{|y|}{L_m} dx \qquad R_a = \sum_{i=1}^{n} \frac{|y_i|}{n}$$

SURFACE ROUGHNESS

a lot of different KIND of parameters

$$R_q = \sqrt{\frac{1}{N}\sum_{i=1}^{N} y_i^2}$$

Related to square deviation



total variation from highest pick to deepest valley !

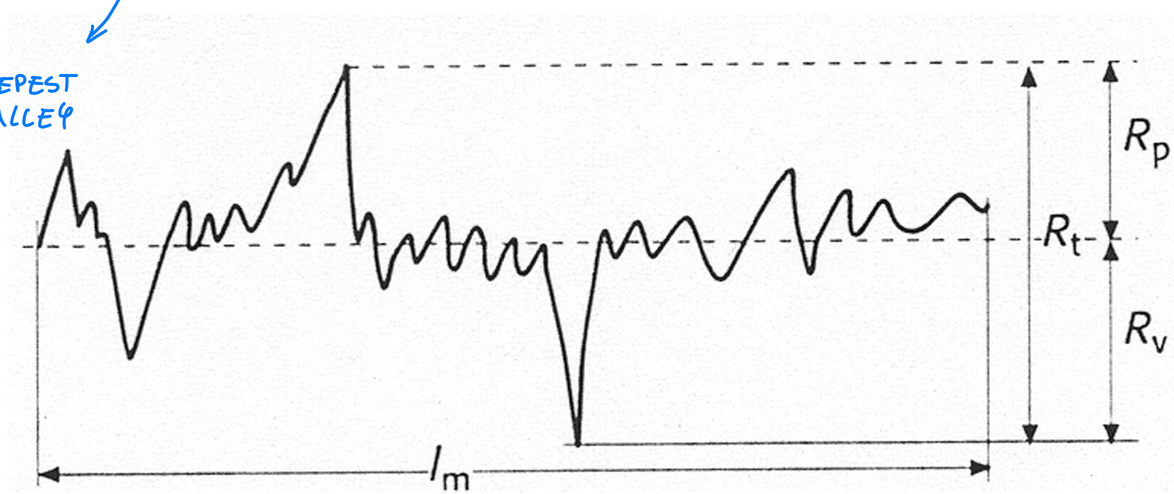
TOTAL (max) Roughness

$$R_t = R_p + R_v$$

DEEPEST VALLEY

HIGHER PROFILE PICK

measure total variation



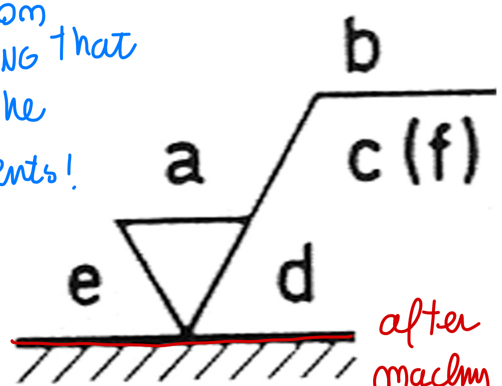
Surface Roughness Specification

Surface texture symbols in engineering drawings

enable us to represent Requirements

SURFACE REQUIREMENTS on Roughness/ other properties

symbol on the DRAWING that represent the Requirements!



after machining surface

a) R_a or roughness grade;

$R_a \leq a$ AFTER Machining

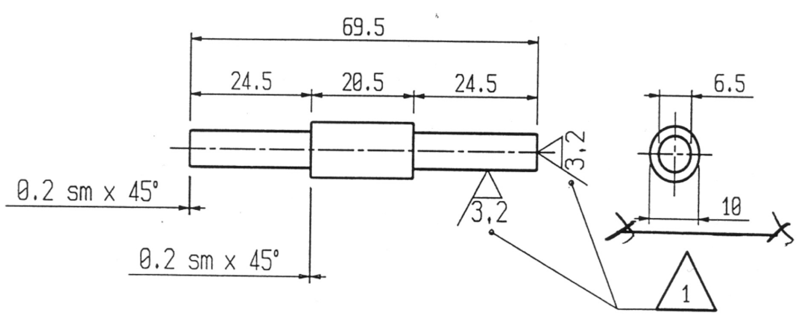
b) waviness;

c) cutoff length;

d) lay symbol;

e) machining allowance;

f) other roughness parameters.



after machining requirements

we can do the same measurements but measuring the SURFACE

Without removing WAVINESS information ⟹ characterize waviness
(similar parameters to Roughness
thanks to more measurement
Using same measuring device)

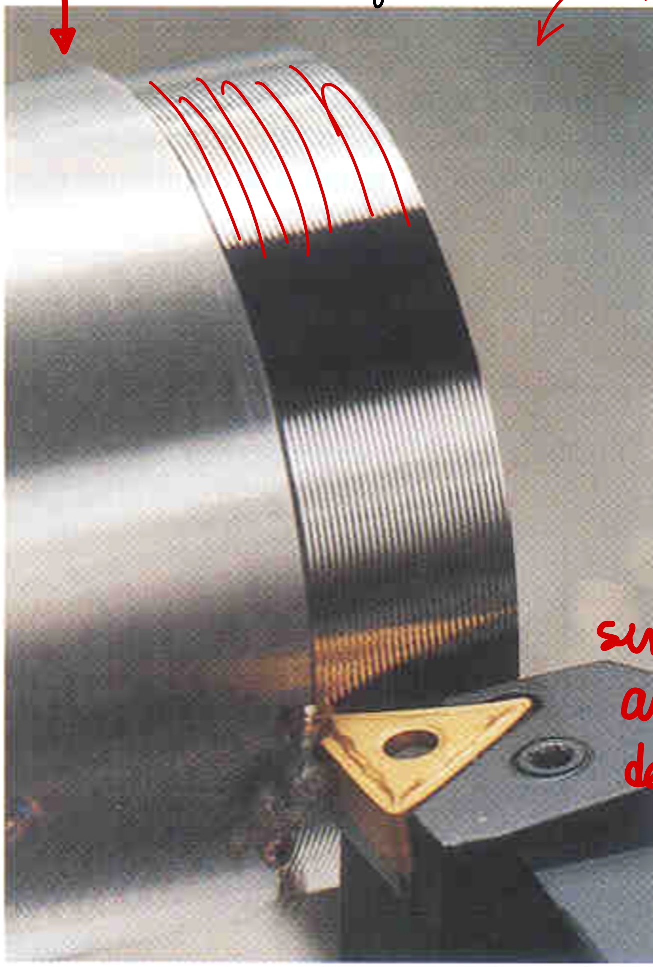
to understand if the choice of Geometry of cutter / process parameters satisfy that requirements...

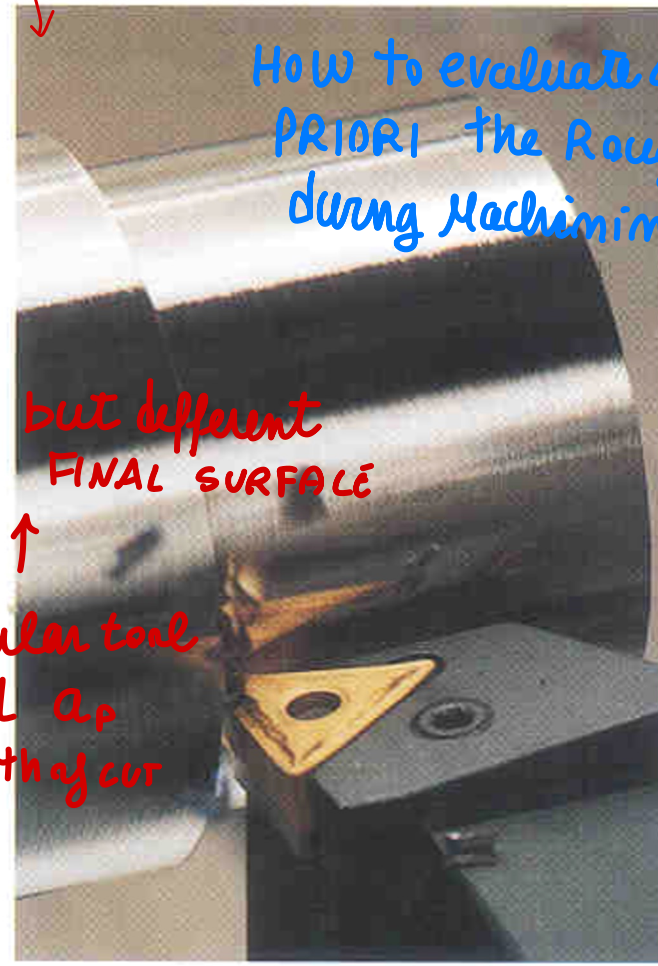
SURFACE ROUGHNESS IN TURNING

We should see what happens!

to understand / satisfy this requirements you should understand what happens during cutting

different surface finish





How to evaluate a PRIORI the Roughness during Machining

but different FINAL SURFACE

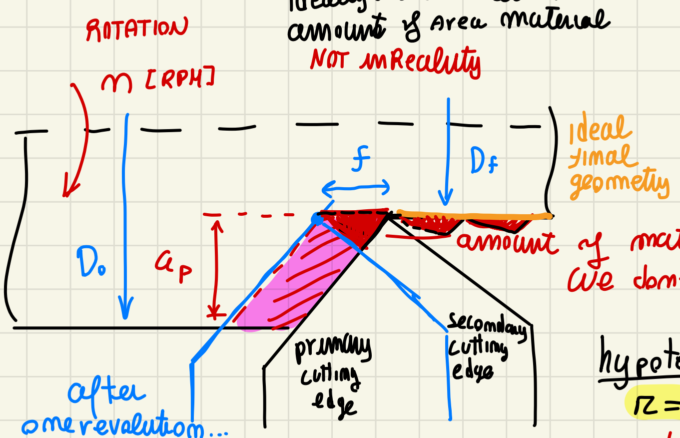
similar tool and a_p depth of cut

Discuss about Roughness on TURNING...

different final surface... **TURNING ROUGHNESS**

consider a classical external cylindrical turning situation

Ideally we cut a certain amount of Area material
NOT in Reality



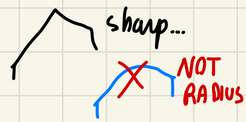
portion of material we suppose to cut, BUT in reality we didn't cut (small amount of material)

↑ using a tool like this one.. without any radius in between the cutting edges

hypotesis $R = 0$

this material NOT removal issue create a profile...

periodic regular profile



h/λ small.. ratio < 50

↓

| Roughness representation |

to measure this surface Roughness

final geometry like:

(x axis as mean axis for the area)

small variation, due to tool interaction with material (NOT due to vibration, ecc...)

so there is a fixed wavelenght λ, and height h (λ related to f) h small also

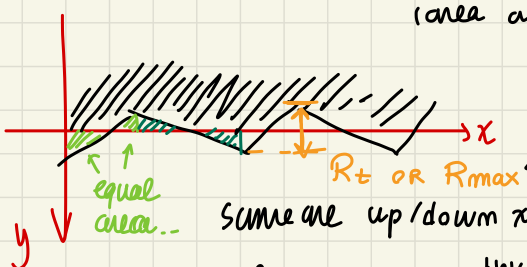
small variations.. ← (surface)
due to interaction of tool with material

e measure Roughness throught deviation y

(we consider the deviation respect material)

Ratio < 50.. we are in a ROUGHNESS of the surface Representation ← $h/\lambda < 50$

(area above / below the line x is the same !)



R_t TOTAL ROUGHNESS

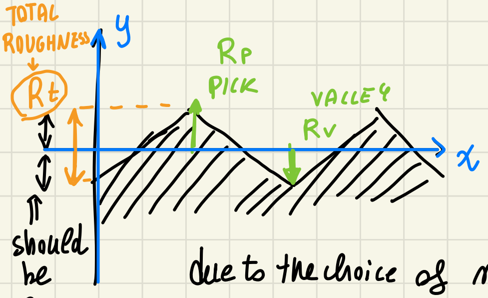
difference between the pick and deepest valley along y

distance costant on the periodic Profile

same are up/down x ..

triangular shape...

↳ in this particular situation of PERIODIC triangular shape profile



to compute the **AVERAGE ROUGHNESS**

$$R_a = \frac{1}{n}\sum_{i=1}^{n} |y_i|$$

should be R_p, R_v

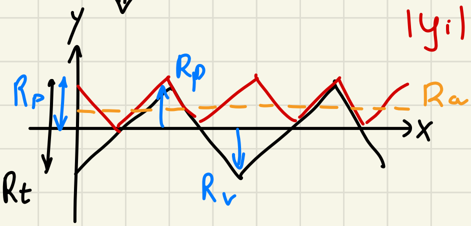
$R_p \simeq R_v$ same deviation

due to the choice of middle surface x... → choose of mid surface so that same area above / below

deviation on pick and valley should be the same

to evaluate R_a → we mirror $y_i < 0 \rightarrow |y_i|$ (to consider the absolute value $|y_i|$)

⇓



average of that profile is on middle

periodic positive profile that we have to AVERAGE to evaluate Ra

Average for triangular profile ⇒ it is in the middle

$$R_a = \frac{1}{2}\left(\frac{1}{2}R_t\right) = \frac{R_t}{4}$$

for this kind of profile (TRIANGULAR)

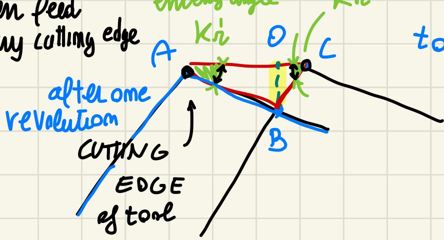
considering this...

in case of cutting work...

(considering our situation, how to estimate Ra)

K_r: angle between feed direction and main cutting edge

K'_r: between feed and secondary cutting edge



tool tip...

$$\begin{cases} \overline{BO} = R_t = (R_p - R_v) \\ \overline{AC} = f \quad (\text{feed}) = AO + OC \end{cases}$$

PICK-VALLEY

after one revolution tip position

⇓ so we can say that:

$$\begin{cases} \dfrac{\overline{OB}}{\overline{AO}} = \tan(K'_r) \\ \dfrac{\overline{OB}}{\overline{OC}} = \tan(K_r) \end{cases}$$

from here...

$$\begin{cases} \overline{AO} = \dfrac{\overline{OB}}{\tan(K'_r)} = \dfrac{R_t}{\tan(K'_r)} \\ \overline{OC} = \dfrac{\overline{OB}}{\tan(K_r)} = \dfrac{R_t}{\tan(K_r)} \end{cases} \Rightarrow \overline{AC} = f = \overline{AO} + \overline{OC}$$

↓

$$f = R_t\left(\cot(K_r) + \cot(K'_r)\right)$$

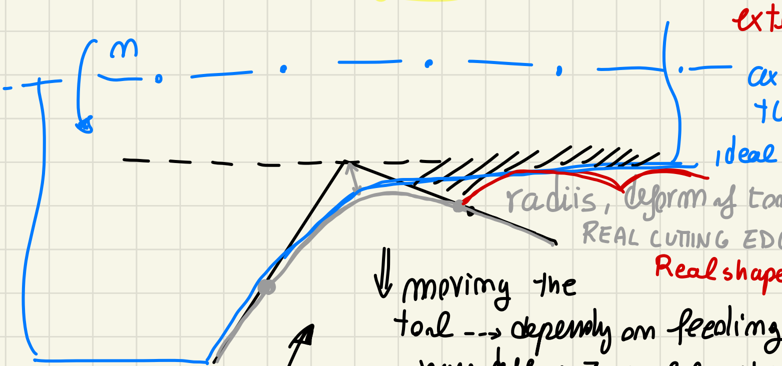
$$R_t = \frac{f}{\cot(K_r) + \cot(K_r')} \cdot [10^3] \quad [\mu m]$$

(to express R_t in μm)

total Roughness

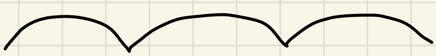
total/average Roughness depends on geometry of the operation and feeding ← (influence the Roughness) $(R_a = R_t/4)$

- **REMOVING HYPOTESIS** → consider $r \neq 0$ (radius, so angle between the two cutting edge)

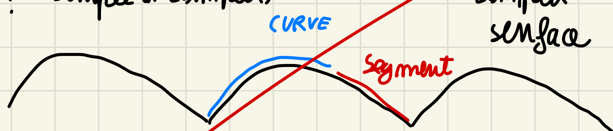


⇓ moving the tool --→ depending on feeding I can have different profile shape of surface! (simple or complex)

Profile as sequence of arcs



if $f << r$ (Just curve)
↑ (small feed respect radius)
We consider that case! better also in real case



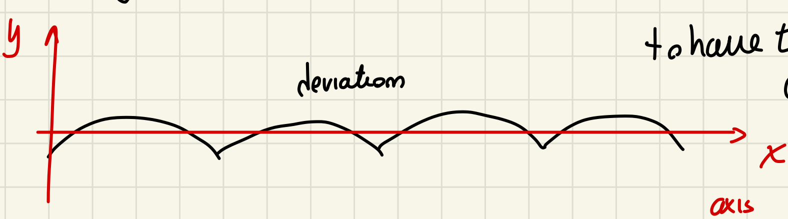
complex surface

if $f > r$ (curve + segment)

AVOID THIS, NOT happen usually, (complex condition)

if big feed, also the linear portion of cutting edge is involved

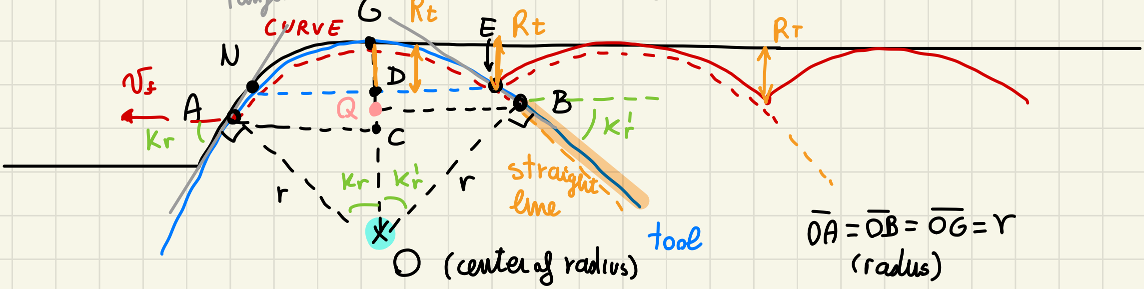
⇓



to have this condition... with $f << r$ and simple arc like shape
↳ What should happen

to have that "desired" easier surface shape (this situation..)

roughness profile is defined by $\widehat{NE}$
while $\widehat{AB}$ is the all arcs of tool

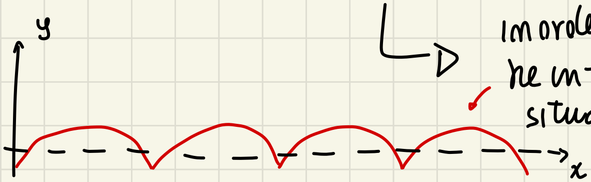


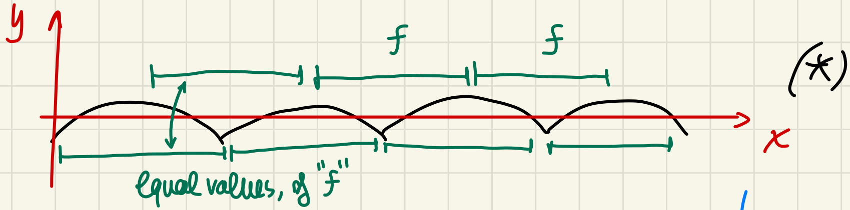
the intersection between the two profile should happen before straight line → to see only the curve part of the cutting edge

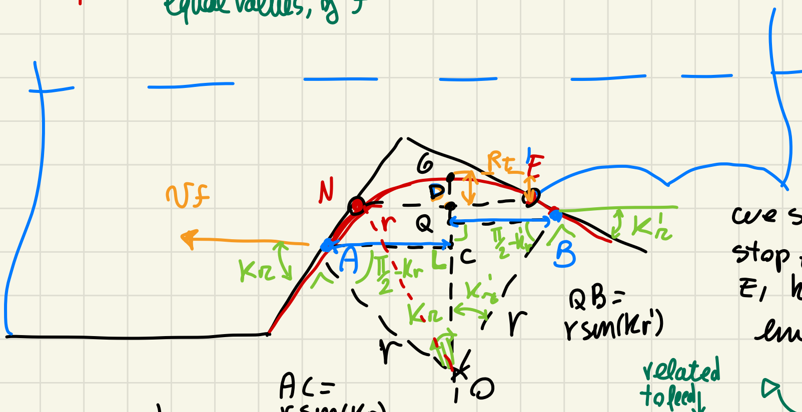
$\widehat{NE}$ as portion of a circle

- Q is projection of B
- C // of A
- D // of N, E

→ in order to be in this ideal profile situation







$OD = \sqrt{ON^2 - ND^2} =$

$= \sqrt{r^2 - (f/2)^2}$ (*)

we should stop feed on E, before straight line EB

QB = r sin(Kr')

AC = r sin(Kr)

related to feed

To have the ideal condition (*)

to evaluate AC, QB, AC // to feed direction (on external cyl. turning)

HYPOTESIS for that profile: (we want) →

$$\left\| \begin{array}{l} \overline{AC} \geq \overline{ND} \\ \overline{QB} \geq \overline{DE} \end{array} \right\|$$

$f = ND + DE$

$\{ND = DE = f/2\}$

because of symmetry of the circle

from this hypotesis..

$$\begin{cases} AC = r \sin(K_r) \\ QB = r \sin(K_r') \end{cases}$$

⇓

Toghether that two conditions, ask for...

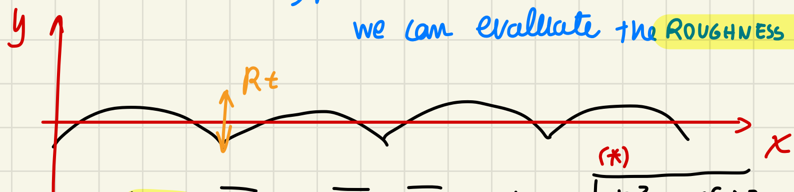
$$\begin{cases} r\sin(K_r) \geq f/2 \\ r\sin(K_r') \geq f/2 \end{cases} \Leftrightarrow \left\| f \leq \min\{2r\sin(K_r); 2r\sin(K_r')\} \right\|$$

Geometrical condition about RADIUS of tool and angle of main/secondary entery

from this hypotesis

IF that hypotesis HOLD and we have that kind of profile...

we can evaluate the ROUGHNESS



$$R_t = \overline{GD} = \overline{OG} - \overline{OD} = r - \overset{(*)}{\sqrt{r^2 - \left(\frac{f}{2}\right)^2}} =$$

approx square root...

$$= r - r\sqrt{1 - \left(\frac{f}{2r}\right)^2} \simeq r - r\left[1 - \frac{1}{2}\left(\frac{f}{2r}\right)^2\right] = \cancel{r} - \cancel{r} + \frac{\cancel{r}}{2}\frac{f^2}{4r^{\cancel{2}}} \Rightarrow$$

MC LAURIN approx

$$\boxed{R_t = \frac{f^2}{8r} \cdot [10^3]} \ [\mu m]$$

to have $[\mu m]$

Total Roughness related to feed^2 and radius of tool (inversal prop. to r)

- when the tool has an $r \neq 0$ and the hypothesis of circle profile are respected!

SO TO EVALUATE

$$\boxed{R_a \simeq \frac{R_t}{4} = \frac{f^2}{32r} \cdot 10^3} \ [\mu m]$$

APPROX

AVERAGE ROUGHNESS

(while on triangular shape was a correct value!) exact

SURFACE ROUGHNESS IN TURNING

- **Theoretical roughness:** roughness that can be computed theoretically starting from the geometrical characteristics of the process.

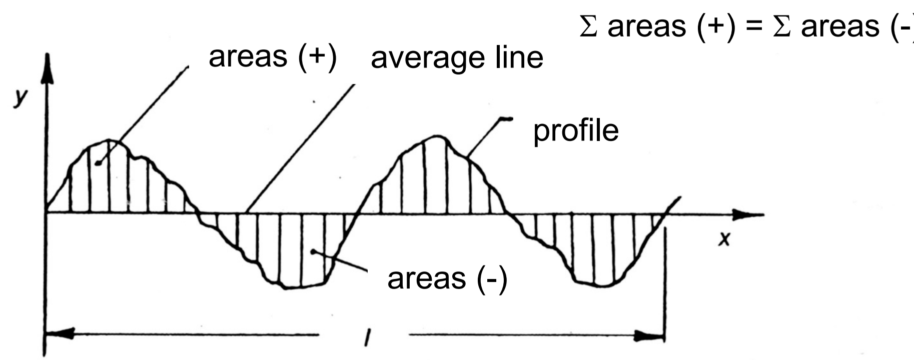
- **Real roughness:** actual roughness of the machined part.

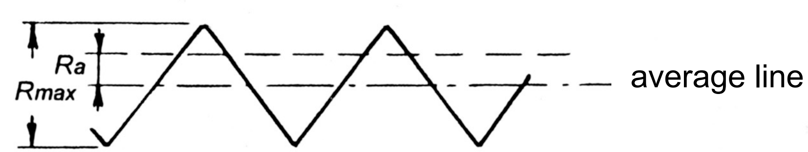
Roughness Parameters

- ***R_{max}*** or ***R_t***: "maximum distance (expressed in μm) between the highest peaks and the deepest valleys".
- ***R_a***: "arithmetic average (expressed in μm) of the absolute values of the deviations *y* of the real profile from the average line".

$$R_a = \frac{1}{l}\int_0^l |y| \cdot dx$$

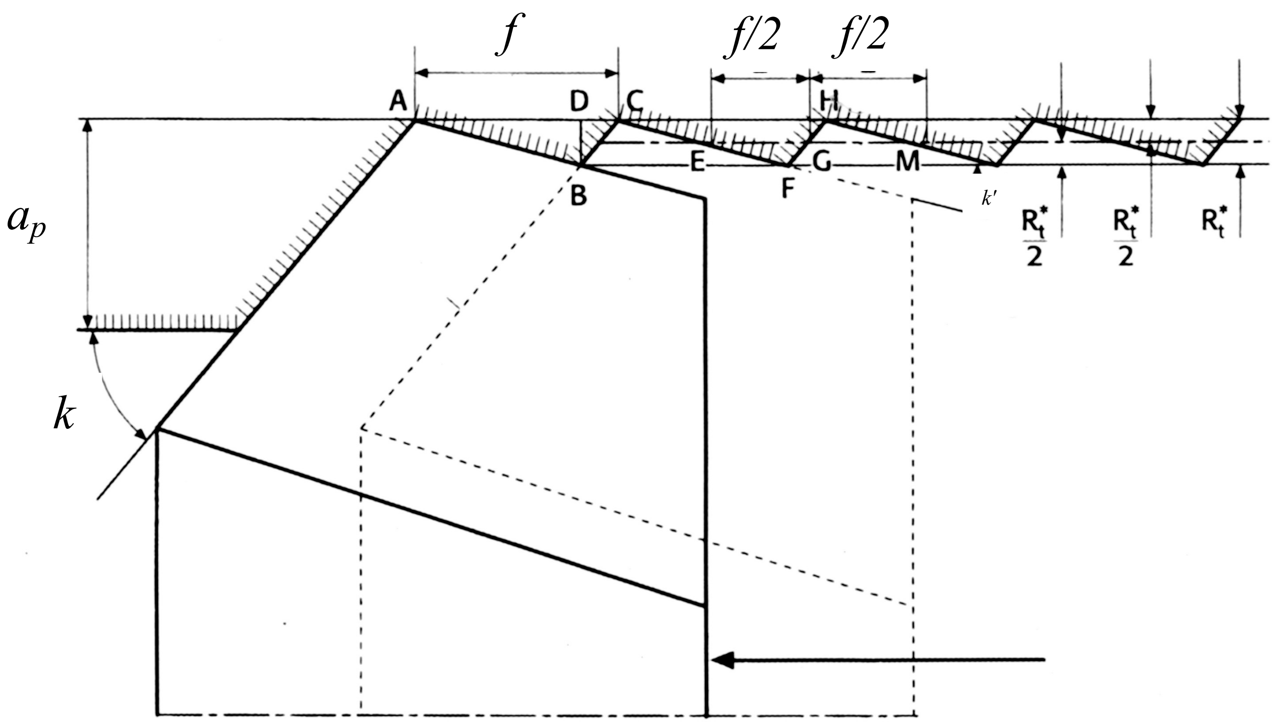
$$R_a \cong \frac{R_{\max}}{4}$$





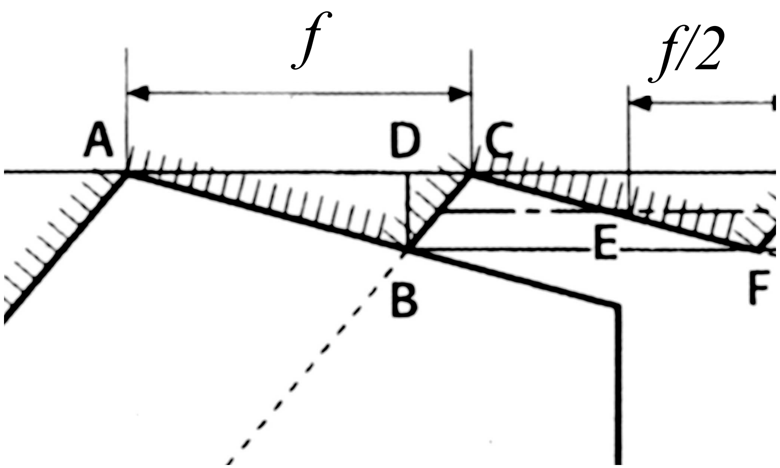
SURFACE ROUGHNESS IN TURNING

Theoretical roughness: tool with zero nose radius



Surface Roughness in Turning

Theoretical roughness: tool with zero nose radius



f is given by:

$$f = \overline{AD} + \overline{DC}$$

$$f = \overline{DB}\left(\text{cotg } k' + \text{cotg } k\right)$$

since:

$$\overline{DB} = R_t = R_{\max}$$

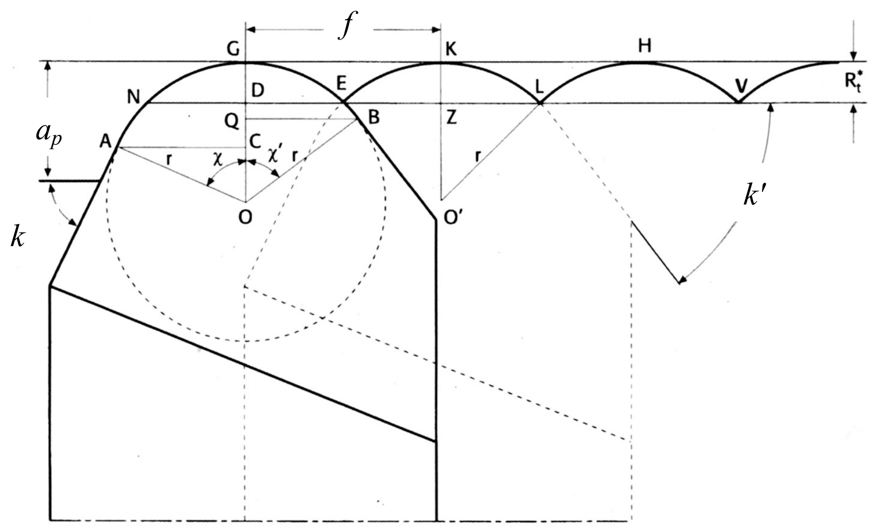
It follows that:

$$R_{\max} = \frac{f}{\text{cotg } k' + \text{cotg } k} \cdot 10^3 \text{ [µm]}$$

$$R_a = \frac{R_{\max}}{4}$$

SURFACE ROUGHNESS IN TURNING

Theoretical roughness: tool with nose radius > 0



since:

$$\overline{ND} \leq \overline{AC}$$

$$\overline{DE} \leq \overline{QB}$$

$$\overline{ND} = \overline{DE} = \frac{f}{2}$$

$$\Rightarrow \quad \frac{f}{2} < r \cdot sin(k') \Rightarrow f < 2r \cdot sin(k')$$

$$\frac{f}{2} < r \cdot sin(k) \Rightarrow f < 2r \cdot sin(k)$$

Then we obtain:

Approx. **Schmaltz** formula

$$R_{\max} = R_t = \overline{OG} - \overline{OD} = \overline{OG} - \sqrt{\overline{ON}^2 - \overline{ND}^2} = \left(r - \sqrt{r^2 - \frac{f^2}{4}} \right) 10^3 \approx \frac{f^2}{8 \cdot r} 10^3 \quad [\mu\text{m}]$$

SURFACE ROUGHNESS IN TURNING

PROPORTION.

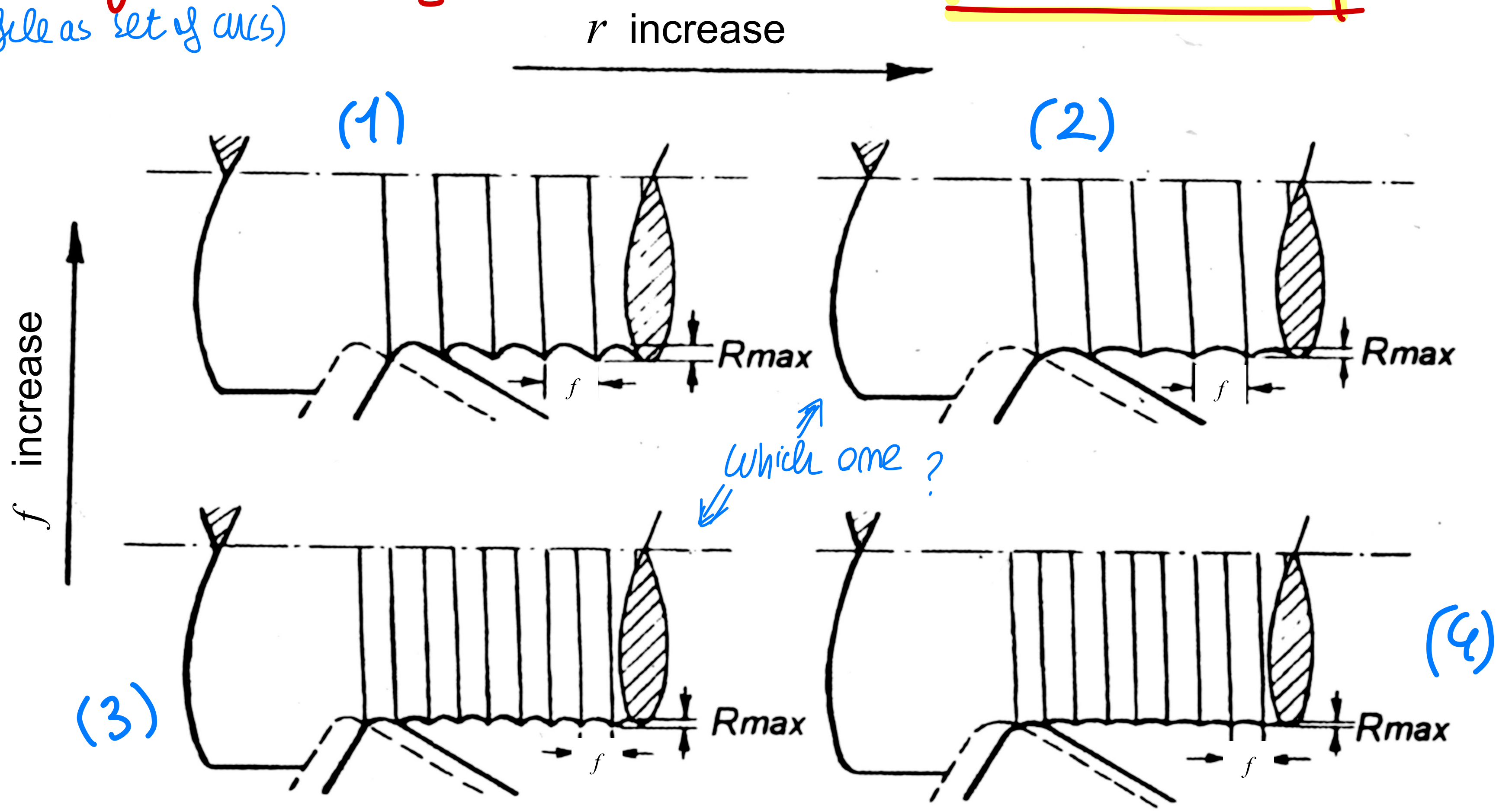
Effects of feed f and nose radius r

$$\boxed{R_a = \frac{1000}{32}\frac{f^2}{r}}$$

$\begin{cases} Ra \propto f^2 \\ Ra \propto 1/r \end{cases}$

Discussing that Result of Ra

(profile as set of arcs)



from the Formula $Ra = \left(\frac{1000}{32}\right)\frac{f^2}{r}$

- $f \uparrow \rightarrow Ra \uparrow$ WORST QUALITY
- $r \uparrow \rightarrow Ra \downarrow$ BETTER QUALITY

by increasing both **(2)**, we have same Ra, but higher (f, r) respect **(3)** with low (r, f)

What is better to choose if we have same Roughness Ra **?**

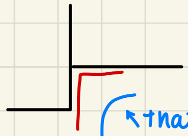
Notice: bigger r means more ROBUST tool (r high good)

- (2) ROBUST TOOL, LESS CUTTING TIME ✓

shape problem: if you have the goal to shape a perfect square element

Notice that $r = 0$ perfectly 90° planar face is impossible

you have small r in good condition



↖ that radius $r\uparrow$ high radius don't allows you to machine it

+ other issues of $r\uparrow, f\uparrow$ **MAIN ISSUE OF (2)**

- $F_c \sim f^{(1-x)}$ PROPORTIONAL

IF $f \uparrow$ (with other constant condition) we require an higher Force! $\Rightarrow F_c \uparrow$

so it impact $P_c \uparrow \rightarrow$ still $P_c \le \eta P_e$? It must remain feasible

- you also have to verify that $K \cdot \frac{F_c L^3}{E J}$ Bending $\uparrow \rightarrow$ more geometrical error!

SO..

you may evaluate $\begin{cases}\text{POSITIVE ASPECTS} \\ \text{NEGATIVE ASPECTS}\end{cases}$ in changing process parameters

OVERALL picture to take right decision → TRADE-OFF

perform TRADE-OFF through KPI objective function to choose BEST SOLUTION

obviously, if feed increases $f\uparrow$ $\Rightarrow$ more Roughness
$\downarrow$
{Worst quality!}

It can vary r or f..
different condition possible..

Same Roughness if
$r\uparrow f\uparrow$
$r\downarrow f\downarrow$
} which best?

(CASE 2)

if

- $f\uparrow$ High feed
given m

$r\uparrow$ High radius

$V_f = m f\uparrow$ + ROBUST TOOL $\rightarrow$ good advantages!

$t_c = \frac{l_c}{V_f}\downarrow$ (faster, less contact time!)

POSITIVE EFFECT!

BUT, ISSUE

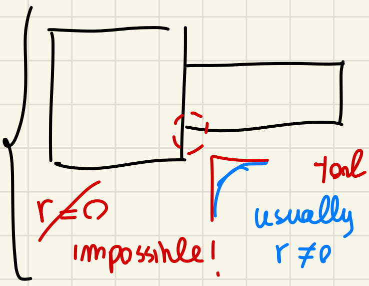
- related to $\uparrow F_c \sim f^{1-x}$ IF $f\uparrow \Rightarrow F_c\uparrow$

HIGHER FORCE required

Disadvantage

- $P_c\uparrow \leq \eta P_e$ We must stay on limit!

- $K\frac{F_c L^3}{EJ}$ bending $\uparrow$
geometrical error increase!



final shape related to radius

f, r? TRADE OFF for chosing
$\downarrow$
We need an OBJECTIVE FUNCTION
$\downarrow$
what I want to optimize!

trade off, respect to

from goal we decide how TO Choice

choices depends on COMPANY STRATEGY goals

$\Downarrow$ Best solution optimization of cutting condition respect
$\Leftarrow$ an objective function